PDS readout

Peak descriptors and trigger primitives

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Tests exposed a host-processing bottleneck

Test: 16 channels, 120 kHz target, noise descriptors.

Host processing	NIC loss	Raw queue loss	Descriptor loss
Callbacks on RX thread	51.91%	—	—
Callbacks on workers	0%	5.55%	—
Separate descriptor workers	0%	4.37%	0%

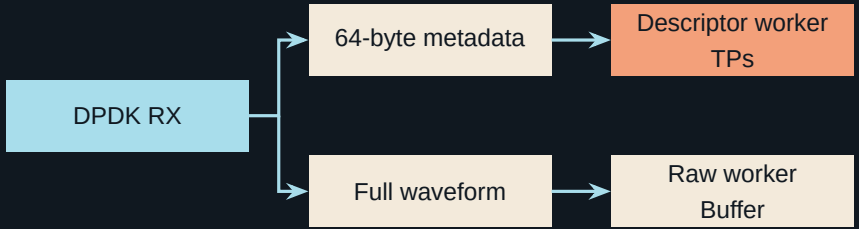
Offload callbacks so RX keeps draining the NIC ring.

Separate descriptors so raw backlog does not block TPs.

Host software changes; NIC firmware unchanged.

Separate short tests; descriptor populations vary. Not a losslessness qualification.

One frame feeds two host paths



Metadata is queued before the raw-buffer callback.

Raw queue saturation need not discard descriptor work.

`dpdklibs`: `IfaceWrapper.cpp`, `FrameWorker.hpp`

`fdreadoutlibs`: `DAPHNEEthFrameProcessor`, `DescriptorProcessor`

Package changes: descriptor TPs and TA handoff

Package	Change
`fddetdataformats`	Format-4 frame: 256 samples + five peak descriptors.
`fdreadoutlibs`	Validate descriptors, map channels, emit native TPs.
`dpdklibs`	Fork descriptor processing before raw buffering.
`appmodel`	Configure raw/descriptor workers and separate TPG.
`triggeralgs`	Session-local prescale fix: PDS type, one-tick samples.

Tested base: fddaq-v5.6.2-a9-1. Source pins: companion notes.

Keep 1024, 256 and full-stream data supported

PROPOSED: fddetdataformats

Expand the version field

Use the packet version to select 1024- or 256-sample decoding.

Format	Decoder
1024 samples	Retain legacy decoding.
256 samples	Decode compact waveforms + peak descriptors.
Full stream	Retain streaming decoding.

Coordinate version encoding between firmware and DAQ.

Run 45392 proved PDS-TA-triggered readout

PDS TPs → PDS TAs → TCs → MLT → readout

DECODED

29 matching 1 ms windows

Every window: all 32 PDS channels across two links + emulated charge.

PDS supplied the triggers

Software captures; TA prescale 100.
Charge trigger generation disabled.

PDS + charge payloads: 29/29 clean

First record: TC metadata status 0x12;
candidate decoded, payloads complete.

Evidence: decoded records + configuration